Capstone Project Proposal: VeriCBAM

Multimodal Evidence Fusion & Decision Support for CBAM Emissions Consistency Assessment

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1. Project Overview & Problem Statement

1.1 The Regulatory & Economic Context

On 1 January 2026, the European Union's Carbon Border Adjustment Mechanism (CBAM) entered its definitive financial enforcement phase under Regulation (EU) 2023/956. Following a two-year transitional reporting phase (October 2023 – December 2025 governed by Commission Implementing Regulation (EU) 2023/1773), European industrial importers—designated as Authorised CBAM Declarants—importing carbon-intensive commodities (Cement, Iron & Steel, Aluminium, Fertilisers, Electricity, and Hydrogen) face strict legal liability. They are required by law to surrender CBAM certificates annually corresponding to the verified embedded direct (Scope 1) and indirect (Scope 2) greenhouse gas emissions (tCO_2e) of their imported goods. These certificates are priced at the weekly average EU Emissions Trading System (EU ETS) auction price of allowances (Directive 2003/87/EC), typically trading between €65 and €90+ per metric tonne of CO_2.

While the Omnibus I Simplification Regulation (EU) 2025/2083 introduced a cumulative 50-tonne annual mass de minimis exemption threshold for small importers, approximately 99% of embedded industrial emissions remain subject to mandatory third-party verification and financial certificate surrender.

1.2 The Industrial Screening Dilemma & Verification Bottleneck

European industrial buyers, customs declarants, and National Competent Authorities (such as Germany's Deutsche Emissionshandelsstelle - DEHSt, France's DGEC, the Netherlands' NEa, and Poland's KOBiZE) face an acute operational and economic bottleneck:

1. The Escalating Financial Cost of Physical Audits:

Under Regulation (EU) 2023/956 (Articles 8 & 9), Commission Implementing Regulation (EU) 2025/2546 (verification principles), Commission Delegated Regulation (EU) 2025/2551 (verifier accreditation), and Commission Implementing Regulation (EU) 2025/2547 (calculation of embedded emissions), physical on-site installation visits represent the regulatory default when claiming actual embedded emissions. Third-party verification costs scale steeply with industrial complexity:

- Simple Units: €5,000 to €15,000 per facility (e.g., standalone scrap electric arc furnace or downstream rolling mill).
- Medium Heavy Industrial Sites: €15,000 to €50,000 per facility (e.g., standalone cement clinker rotary kiln).
- Complex Integrated Heavy Industry (The VeriCBAM Target Focus): €50,000 to €150,000+ per site visit (industry estimates; GMK Center, 2024; EC SWD(2021) 643 final) for integrated Blast Furnace–Basic Oxygen Furnace (BF-BOF) steelworks and multi-kiln cement mega-complexes. When factoring in international auditor deployment to third countries (e.g., India, China, Turkey, Brazil, Ukraine), forensic chemical assay of raw meal/coke, and local logistics, comprehensive compliance burdens impose substantial friction on international supply chains.

2. The "Arithmetic of Scarcity" & Administrative Timeline Delays:

A severe structural market failure compounds this cost burden:

- Demand vs. Supply Deficit: An estimated ~12,000 EU import entities are seeking authorized CBAM declarant status across tens of thousands of supplying installations worldwide. Yet historically, only ~403 verification bodies held EU ETS accreditation under the Accreditation and Verification Regulation (AVR - Commission Implementing Regulation (EU) 2018/2067) across Europe; dedicated CBAM verifier accreditations under Commission Delegated Regulation (EU) 2025/2551 roll out gradually from late 2026.
- Extended Verification Lead Times: A complete physical site verification cycle takes 3 to 6 months from initial engagement, monitoring plan reconciliation, on-site physical walk-through, to final report sign-off.

- The Punitive Alternative (Default Value Mark-Up): If an overseas installation cannot secure verified data in time, importers are forced to surrender CBAM certificates calculated via the European Commission's default emissions values, which carry punitive administrative mark-ups under Implementing Regulation (EU) 2025/2547:

Markup = +10% (2026) → +20% (2027) → +30% (from 2028).

For an integrated steel mill exporting 500,000 tonnes of steel, this default surcharge imposes an unrecoverable €5,000,000 to €20,000,000+ penalty.

3. Three Critical Evasion & Technical Inconsistency Vectors:

- Vector A — Selective Quantitative Understatement: Deliberately under-reporting fuel consumption or raw material calcination by 10%–50% to evade certificate surrender liability while maintaining normal physical manufacturing output.
- Vector B — Stoichiometric Floor & Boundary Manipulation: Declaring specific embedded emissions below the physical chemical conservation-of-mass limits required to reduce iron ore or calcine limestone, or exporting high-calorific blast furnace process gases across legal fence lines to adjacent power plants without allocating them to product emissions.
- Vector C — Process Route Misclassification ("Greenwashing"): Claiming that high-carbon primary steel (produced via integrated BF-BOF, ≈ 1.85 t CO_2/t) was manufactured via a low-carbon secondary route (Scrap-EAF, ≈ 0.08 t CO_2/t direct Scope 1).

1.3 The VeriCBAM Objective & Scope Boundaries

VeriCBAM is an explainable, multimodal industrial intelligence and decision-support system designed to screen and assess the consistency of self-reported CBAM declarations.

Rather than claiming to replace accredited legal certification bodies or directly measure facility CO□ emissions from space, VeriCBAM functions as an evidence-fusion auditor copilot:

- It establishes physical plausibility via first-principles chemical engineering process bounds and Best Available Techniques (BAT) benchmarks.
- It evaluates operational activity consistency via Copernicus Sentinel-5P TROPOMI tropospheric NO_2 atmospheric plume contrast.
- It compares behavior against facility historical fingerprints and multi-pollutant co-emission ratios (NO_x/CO_2 and SO_x/CO_2) from verified industrial registries.
- It integrates these heterogeneous data streams through a weighted Bayesian evidence-fusion engine (Model E1) and a facility-grouped supervised triage model (Model E2) that quantify risk alongside explicit confidence and uncertainty intervals.
- It delivers an explainable reasoning graph and audit dossier for human decision-makers, including a standalone pre-screening tool for importers and overseas suppliers.

Methodological Scope & Regulatory Proxy Qualification:

Because official third-country CBAM declarations are legally confidential during the transitional phase, VeriCBAM's empirical benchmark utilizes authentic European facility-level Scope 1 emissions from the European Environment Agency (EEA E-PRTR / IED v16 database) as an independent regulatory proxy rather than product-level CBAM embedded intensities (EU Implementing Regulation 2025/2547). The system is positioned strictly as a verification and triage decision-support tool, preserving human auditor authority under EU administrative law.

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2. Project Objectives & Technical Hypotheses

2.1 Core Problem Statement & Overarching Objective

Can heterogeneous physical, observational, historical, and multi-pollutant evidence be systematically fused into an explainable decision-support model that identifies controlled CBAM emissions inconsistencies more reliably than isolated single-modality sources?

2.2 Technical Inquiries & Working Hypotheses

- Objective 1 (Physical Plausibility & Stoichiometric Floors):

Technical Inquiry: How accurately can first-principles chemical engineering mass balances (limestone calcination and iron reduction) establish conservative lower bounds on declared emissions without relying on historical facility data?

Hypothesis H_1: Stoichiometric mass-balance floors will deterministically detect gross understatements (≥ 30%) and route misclassifications with high precision and low false alarms on applicable facilities, but will remain blind to moderate understatements (10%–20%) that remain above the physical floor.

- Objective 2 (Earth Observation Activity & Atmospheric Limits):

Technical Inquiry: Can Copernicus Sentinel-5P TROPOMI tropospheric NO_2 column density serve as a quantitative operational activity proxy for heavy industry, and what are its physical limitations?

Hypothesis H_2: Due to coarse spatial footprint (5.5 × 3.5 km), regional background advection, wind dispersion, and persistent cloud obscuration, satellite trace-gas observations alone cannot reliably quantify annual plant-level CO_2 emissions (yielding low discriminative power when uncoupled from ground activity), but provide valuable qualitative operational gating when clear-sky conditions exist.

- Objective 3 (Historical Behavior & Multi-Pollutant Fingerprinting):

Technical Inquiry: Can facility-specific multivariate historical profiles and co-pollutant combustion ratios (NO_x/CO_2) detect selective CO_2 under-reporting in newly declared reporting periods?

Hypothesis H_3: High-temperature combustion couples NO_x and CO_2 formation; under controlled conditions where declared CO_2 is selectively understated while permit-regulated NO_x remains unperturbed, this will induce a statistically detectable divergence in the leave-one-year-out co-pollutant ratio, independent of remote sensing.

- Objective 4 (Multimodal Evidence Fusion vs. Single Modalities):

Technical Inquiry: Does fusing physical, observational, historical, and co-pollutant evidence streams yield superior discrimination, probability calibration, and false-positive control compared to isolated modalities?

Hypothesis H_4: While historical trajectory alone is hypothesized to achieve strong statistical anomaly discrimination, combining it with stoichiometric boundaries and confidence gating in a weighted Bayesian framework (Model E1) will provide an ultra-low false-positive rate on authentic compliant plants (≤ 1%) and full legal explainability, while supervised logistic fusion (Model E2) will optimize overall probability calibration.

- Objective 5 (Uncertainty Quantification & Missing Data Handling):

Technical Inquiry: How can an evidence-fusion engine systematically represent observational uncertainty (e.g., cloud cover, unobserved capacity, missing historical years) without generating overconfident false alarms?

Hypothesis H_5: A formal evidence confidence factor C_i in [0, 1] combined with an explicit `Insufficient Evidence` abstention gate ($C_{overall} < 0.50$) will provide an architectural safeguard against premature adverse audit findings, allowing the system to abstain rather than generating speculative accusations when observational confidence is low.

- Objective 6 (Explainability & Dual-Stage Regulatory Triage):

Technical Inquiry: Can an automated system generate auditable, evidence-grounded dossiers that satisfy the strategic and risk analysis requirements of Commission Implementing Regulation (EU) 2025/2546?

Hypothesis H_6: A dual-stage architecture—deploying supervised classification (Model E2) for high-throughput customs port triage and white-box Bayesian log-odds reasoning (Model E1) for accredited verifier audit dossiers—will support the strategic and risk analysis principles of Commission Implementing Regulation (EU) 2025/2546 while maintaining auditable evidential trails.

3. System Architecture & Mathematical Formulations

System Architecture & Dataflow: End-to-end multimodal pipeline linking document ingestion to physical bounds, remote sensing, and dual-stage Bayesian/logistic inference. (Interactive visualization available in Streamlit UI Tab 1)

3.1 Chemical Engineering Stoichiometry (Conservation of Mass)

Physical mass-balance floors represent deterministic boundaries rooted in chemical stoichiometry and European Commission Best Available Techniques (BAT):

A. Cement Clinker Calcination & BAT Thermal Floor

Thermal decomposition of calcium carbonate ($CaCO_3$, molar mass 100.0869 g/mol) into quicklime (CaO, 56.0774 g/mol) and carbon dioxide (CO_2, 44.0095 g/mol):

$$\mathbf{CaCO_3 \text{—[}\ \Delta H = +178\ kJ/mol\ \text{]}\rightarrow CaO + CO_2\ (gas)}$$

$$\gamma_{CaO} = (M_{CO_2} / M_{CaO}) = (44.0095 / 56.0774) = 0.78480 \text{ t } CO_2\text{/t CaO}$$

Standard Portland cement clinker requires a minimum mass fraction $f_{CaO} \geq 0.63$ (central 0.65), yielding an inescapable chemical calcination floor:

$$EF_{calc,\,min} = 0.63 \times 0.78480 \approx 0.495 \text{ t } CO_2\text{/t clinker} \quad \text{(Central: 0.525 t } CO_2\text{/t clinker)}$$

Adding the Best Available Techniques (EU CLM BREF 2013) minimum thermal fuel combustion for a state-of-the-art 5-stage preheater/precalciner dry kiln (2.9–3.3 GJ/t clinker) establishes:

$$EF_{thermal,\,min} = 0.275 \text{ t } CO_2\text{/t clinker}$$

$$E_{floor,\,clinker} = P_{clinker} \times (0.495 + 0.275) - E_{CCUS,\,verified} = 0.770 \text{ t } CO_2\text{/t clinker} - E_{CCUS}$$

B. Primary & Secondary Steelmaking Route Floors

- Integrated Blast Furnace – Basic Oxygen Furnace (BF-BOF):

Carbothermic reduction of hematite ore (Fe_2O_3 + 3C arrow 2Fe + 3CO arrow $3CO_2$) establishes a reduction minimum of 1.182 t CO_2/t hot metal. Including blast furnace thermal enthalpy balance (300–360 kg coke/t), sintering/coking, and up to 15%–20% scrap dilution in the BOF converter:

$$EF_{floor,\,BF\text{-}BOF} = 1.350 \text{ t } CO_2\text{/t crude steel} \quad \text{(European industrial benchmark: 1.75–1.95 t/t)}$$

- Direct Reduced Iron – Electric Arc Furnace (DRI-EAF, Natural Gas Shaft):

$$EF_{floor,\,DRI\text{-}EAF} = 0.550 \text{ t } CO_2\text{/t crude steel (Direct Scope 1)}$$

- Secondary Scrap – Electric Arc Furnace (Scrap-EAF):

Melting 100% recycled ferrous scrap (emissions from graphite electrode wear and burners only):

$$EF_{floor,\,Scrap\text{-}EAF} = 0.040 \text{ t } CO_2\text{/t crude steel (Direct Scope 1)}$$

C. Conservative Utilization Proxy & Transferred Process Gas Boundary Rule

When auditing against external registry capacity (rather than audited production), VeriCBAM evaluates the conservative floor at minimum plausible utilization ($\eta_{min} = 0.50$):

$$E_{floor,\,conservative} = C_{nameplate} \times 0.50 \times EF_{floor,\,route}$$

D. The Real-World Industrial Boundary Problem (Why Textbook Chemistry Fails Without System Modeling)

In academic literature, mass-balance stoichiometry is often treated as a trivial textbook exercise. In industrial practice, however, heavy industrial facilities are not isolated closed systems:

- Off-Gas Transfers Across Fence Lines: In modern integrated blast furnace steelworks, carbothermic reduction produces enormous volumes of high-energy Blast Furnace Gas (BFG) and Coke Oven Gas (COG). In major industrial clusters, these process gases are routinely piped across legal property lines to external utility power stations or adjacent district heating networks. Consequently, the direct Scope 1 stack emissions of the steelmaking installation appear deceptively low, because a substantial portion of the chemical carbon leaves the legal perimeter in gaseous form to generate off-site electricity.
- The Naive Model Failure: If an audit system naively evaluates the steel plant using standard textbook stoichiometry alone, it inevitably triggers false alarms on compliant facilities. VeriCBAM resolves this by incorporating cross-registry system boundary mapping: for facilities with documented off-gas export infrastructure (e.g., ArcelorMittal Bremen, Eisenhüttenstadt, Taranto, HKM Duisburg, Tata Steel IJmuiden), the stoichiometric stream sets `stoich_applicable = False` and explicitly abstains ($\hat{p}$ = NaN) until complete site energy balances are provided, eliminating false accusations against compliant producers.

E. The Structural Blind Spot of Physical Basements

Stoichiometric process bounds establish a deterministic lower basement (e.g., 1.350 t CO_2/t for BF-BOF steel; 0.770 t CO_2/t for cement clinker). However, physical floors have an inherent structural blind spot:

- If an integrated blast furnace normally operates at 1.90 t CO_2/t, and the operator under-reports its emissions as 1.55 t CO_2/t, stoichiometry alone is completely blind to this evasion, because 1.55 > 1.35.

- Physical boundaries are mathematically incapable of detecting moderate understatements that remain above the theoretical minimum. To close this gap, an audit system must fuse physical limits with longitudinal statistical baselines and multi-pollutant combustion profiles.

3.2 Multi-Pollutant Combustion Ratio (NO_x/CO_2)

Within-facility high-temperature combustion couples fuel/thermal NO_x and CO_2 emissions (median empirical CV = 11.1% across European plants). For facility i in year t, VeriCBAM calculates the leave-one-year-out ratio $R_{i,-t} = \text{Median}_{s \neq t}(E_{NO_x,i,s} / E_{CO_2,i,s})$ and robust scale $\sigma_{R,i,-t}$. A sharp increase in the observed ratio indicates that declared CO_2 dropped without an operational reduction in combustion activity.

A. The Multi-Pollutant Regulatory Invariant Dilemma

The inclusion of co-pollutant ratios addresses the risk of data falsification through a structural regulatory invariant:

- Heavy industrial installations are already subject to continuous environmental monitoring for toxic local air pollutants (NO_x, SO_2, particulate matter) under municipal and national operating permits (such as the EU Industrial Emissions Directive, national air quality laws, or automated continuous emission monitoring systems - CEMS).
- While an operator faces a strong financial incentive to depress declared CO_2 under trade carbon mechanisms, they face immediate legal penalties and operational shutdowns if they tamper with locally mandated air pollution permits.
- If an operator selectively under-reports CO_2 to customs authorities while their local environmental permits reflect normal high-temperature combustion NO_x, the co-pollutant ratio diverges sharply. Reconciling false carbon declarations against separate, independently enforced air quality registries creates an effective deterrence mechanism.

3.3 Longitudinal Historical Trajectory (One-Sided Pooled MAD Scale)

For facility i in year t, longitudinal log-ratio drift is evaluated against the historical median $M_{i,-t} = \text{Median}_{s \neq t}(\ln E_{CO_2,i,s})$ and scaled by the cohort-wide pooled Median Absolute Deviation ($\sigma_{pooled} = 0.0911$ across 177 plant-years):

$$z_{drop,i,t} = \max(0, -((\ln E_{declared,i,t} - M_{i,-t}) / \sigma_{pooled}))$$

The one-sided operator max(0, -z) ensures that production expansions ($E_{declared} > M_{i,-t}$) receive $z_{drop} = 0$ and are never penalized.

3.4 Earth Observation Activity (Copernicus Sentinel-5P TROPOMI)

Tropospheric NO_2 column density (mol/m^2, Level-2 product filtered at `qa_value >= 0.50`) is extracted within a 5 km facility radius (V_{target}) and compared to a 15–30 km background ring (V_{bg}):

$$z_{sat} = ((V_{target} - V_{bg}) / (1.4826 \times MAD_{bg})), \qquad C_{sat} = \max(0, 1 - f_{cloud})$$

A. Physical & Atmospheric Realities of Spaceborne Observations

A common misconception in carbon accounting is that spaceborne remote sensing can directly meter annual factory emissions. VeriCBAM incorporates satellite data with explicit physical and atmospheric qualifications:

- Spatial Footprint: Sentinel-5P TROPOMI pixels (5.5 × 3.5 km) encompass entire industrial parks, nearby transport arteries, and surrounding residential heating, diluting point-source signals.
- Atmospheric Dispersion & Lifetime: Tropospheric NO_2 is a short-lived photochemical tracer heavily influenced by planetary boundary layer height, wind advection, and hydroxyl radical chemistry, introducing significant meteorological variance.
- Persistent Cloud Obscuration: In temperate industrial regions, cloud decks obscure optical spectrometers for substantial portions of the year, providing sporadic clear-sky overpass snapshots rather than continuous monitoring.
- The True Role of Satellite Data: Spaceborne observation functions effectively as an auxiliary operational activity gate—verifying whether a facility was physically active, cycling, or completely idle—rather than an annual emissions meter.

3.5 Weighted Bayesian Evidence Fusion (Model E1)

$$\text{logit } P(H \mid E) = \text{logit } P(H) + \Sigma_{(k=1..6)}\ w_k \cdot C_k \cdot \ln(LR_k)$$

- Baseline Prior: P(H) = 0.08 (logit P(H) = -2.4423).
- Evidential Weights: Route Match (w=2.00), Stoichiometric Floor (w=1.50), Multi-Pollutant Ratio (w=1.20), Historical Drop (w=1.00), Satellite Plume (w=1.00), Registry Capacity (w=0.80).

- Monte Carlo Uncertainty & Abstention Gating: 500 log-likelihood perturbation draws generate 95% credible intervals. If overall weighted confidence $C_{overall} < 0.50$, the system outputs `Insufficient Evidence`.

3.6 Supervised Triage Classifier (Model E2)

An L_2-regularized Logistic Regression model trained on standardized multimodal feature vectors via 5-Fold `GroupKFold(groups=facility_id)`, ensuring zero facility or spatial data leakage between folds.

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4. Empirical Datasets: Reference Observations & Graded Benchmark

4.1 Layer A: Real-World Reference Cohort (EEA E-PRTR / IED v16)

- Dataset Provenance: Official European Pollutant Release and Transfer Register (E-PRTR) & Industrial Emissions Directive (IED) integrated database v16.0 (DOI: `10.2909/657ac3cb-affa-4295-a4a9-27b4f539adab`).
- Cohort Structure: Exactly 30 premier European industrial installations (15 Cement, 15 Steel) spanning 9 EU Member States (Germany, Poland, Austria, France, Netherlands, Italy, Slovakia, Greece, Sweden, Denmark).
- Observation Completeness: Exactly 177 authentic facility-year observations across 2018–2023. Three facility-years are missing due to documented blast furnace relinings (e.g., ArcelorMittal Bremen BF2 in 2020) and are left unimputed.
- Biomass Data Sparsity Finding (Step 11): In the official EEA dataset, `co2_excl_biomass_tonnes` is missing (`NaN`) for 144 of 177 facility-years (81.4% missingness), with only 33 non-null records. Total direct `Carbon dioxide (CO2)` is therefore the only complete, verified pan-European regulatory series.

4.2 Verified Technology & Capacity Registry

Constructed from the Global Energy Monitor (GEM) Global Steel & Cement Trackers, corporate sustainability reports, and official national environmental operating permits (e.g., Bezirksregierung Düsseldorf, Arpa Puglia, GIOŚ Poland), documenting:

- Exact process route, clinker/crude steel nameplate capacities, and kiln configurations.
- Verified historical operating utilization bounds (η in [0.50, 0.95]).
- Transferred process gas flags identifying installations exporting blast furnace gas across legal boundaries.

4.3 Layer B: Primary 325-Case Graded Evaluation Benchmark

To rigorously evaluate model performance in the absence of public non-EU fraud datasets, VeriCBAM introduces a curated 325-case graded evaluation suite:

1. 177 Authentic Real-Year Negatives: Unperturbed historical facility-years (y in [2018, 2023]) capturing natural operating variance ($\sigma_{pooled} = 0.0911$).
2. 140 Graded Understatements: Synthetic understatements generated across 5 distinct perturbation magnitudes:

δ in {5%, 10%, 20%, 30%, 50%}, (28 cases per magnitude)

testing detection from subtle deviations (δ = 5%) to severe fraud (δ = 50%).

3. 8 Route Misclassifications: Cross-route anomalies (e.g., claiming Scrap-EAF emissions for integrated BF-BOF works; Hamborn excluded due to zero crude steel capacity).

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5. Proposed Experimental Evaluation Design & Validation Plan

5.1 Comparative Architecture Matrix (Candidate Models)

To validate the technical hypotheses and project objectives, the project proposes benchmarking six distinct statistical and physical paradigms:

Candidate Model	Statistical Paradigm	Input Data Streams	Planned Evaluation Role	Expected Performance Profile
Model A: Naive Prior	Constant Uninformative Baseline	None (P = 0.08)	Null Reference	Theoretical lower bound (ROC-AUC ≈ 0.50).
Model B: Stoichiometry Alone	First-Principles Process Bounds	Tech Registry + Chemical Factors	Hard Physical Veto	High precision on severe violations; blind to moderate deviations.

Candidate Model	Statistical Paradigm	Input Data Streams	Planned Evaluation Role	Expected Performance Profile
Model C: Satellite Alone (S-5P)	Atmospheric Plume Contrast	Sentinel-5P Level-2 Tropospheric NO_2	Auxiliary Presence Check	Qualitative operational presence; limited annual CO_2 resolution.
Model D: Historical Alone	Robust Longitudinal Anomaly Score	EEA E-PRTR 6-Year Panel	Historical Baseline	Strong anomaly discrimination based on individual facility trends.
Model E1: Expert Bayesian Fusion	Weighted Log-Odds + Confidence Gate	All 5 Evidence Streams Fused	Verifier Audit Dossier	White-box interpretability, near-zero authentic false alarms.
Model E2: Learned Logistic Fusion	Supervised GroupKFold Classification	Standardized Multimodal Features	Customs Triage Engine	Optimized probability calibration and ranked prioritization.

5.2 Graded Evaluation Benchmark Protocol (Layer B)

Because authentic third-country fraud datasets are legally protected and confidential during the transitional phase of CBAM, the framework will be validated on a curated 325-case graded evaluation benchmark:

1. 177 Authentic Real-Year Negatives: Unperturbed facility-years (2018–2023) across 30 European industrial complexes from official EEA E-PRTR reporting, reflecting natural operating variance ($\sigma_{pooled} = 0.0911$).
2. 140 Graded Controlled Understatements: Synthetic understatements generated across 5 distinct perturbation magnitudes:

δ in {5%, 10%, 20%, 30%, 50%}, (28 cases per magnitude)

systematically testing model sensitivity from subtle reporting drift ($\delta = 5\%$) to severe fraud ($\delta = 50\%$).

3. 8 Process Route Misclassifications: Cross-route anomalies (e.g., declaring Scrap-EAF emissions for integrated BF-BOF works).

5.3 Planned Evaluation Metrics & Statistical Validation

The candidate models will be evaluated across three complementary statistical dimensions:

1. Discrimination Performance: Evaluated via Receiver Operating Characteristic (ROC-AUC) and Precision-Recall AUC (PR-AUC). To account for intra-facility correlation across multi-year observations, 95% confidence intervals will be estimated using 1,000 facility-clustered bootstrap resamples.
2. Probability Calibration & Reliability: Assessed using the Brier Score and Expected Calibration Error (ECE) across reliability diagrams to ensure posterior probabilities accurately reflect empirical inconsistency risk.
3. Operational Regulatory Triage Metrics:

- Authentic False Positive Rate (FPR): Measuring false alarm rates on unperturbed compliant facilities to avoid wasting scarce verification resources.
- Recall at Fixed False Alarm Budgets: Comparing all candidate models at standardized operational thresholds (FPR = 5% and FPR = 10%).
- Auditor Resource-Constraint Simulation: Simulating customs inspection capacity under realistic constraints (inspecting only the top 10%, 20%, or 30% highest-risk declarations) and measuring Precision@k and Triage Lift over random auditing.

5.4 Planned Ablation Studies & Sensitivity Analyses

- Stream Ablation Protocol: Systematically removing individual evidence streams (stoichiometry, historical drift, multi-pollutant ratios, satellite contrast) to quantify the marginal contribution of each modality.
- Prior & Weight Sensitivity: Evaluating the Bayesian engine across varying prior odds (P(H) in [0.01, 0.20]) and evidential weight perturbations via Monte Carlo simulation.
- Capacity Utilization Sensitivity: Assessing the robustness of stoichiometric floors under varying conservative utilization bounds (η in [0.40, 0.60]).

5.5 Proposed Success Criteria

The project execution will be evaluated against the following primary success criteria:

1. Statistically Significant Triage Lift: Supervised and Bayesian models must achieve at least a 2.0× precision lift over random inspection in the top 20% risk tier.
2. False Alarm Suppression: The white-box Bayesian architecture (Model E1) must suppress false positive alarms on authentic compliant facilities to below 1.0% at default thresholds.
3. Legal Defensibility & Explainability: The system must generate fully transparent, auditable breakdown dossiers satisfying the strategic risk assessment requirements of Commission Implementing Regulation (EU) 2025/2546.

6. Project Work Plan, Gantt Chart & Risk Management

Capstone Milestone Roadmap: Sequential sprint progression from co-pollutant cohort ingestion to proposal approval, probability calibration, UI cockpit delivery, and January 2027 defense.

6.1 Technical Risk Management Matrix

Identified Risk	Severity	Likelihood	Impact on Project	Proactive Mitigation Strategy
Persistent Cloud Obscuration in S-5P	High	High	Overpass data gaps over North/Central Europe.	Formal clear-sky QA filter (`qa_value >= 0.50`), dynamic confidence gating ($C_{sat} = 1 - f_{cloud}$), and explicit `Insufficient Evidence` abstention gate.
Transferred Process Gases in Steelworks	High	High	False physical violation flags on BF-BOF plants exporting gas to off-site power plants.	Verified GEM/permit boundary mapping setting `stoich_applicable = False` for transferred-gas plants; conservative floor evaluated at 50% capacity.
Biomass Emission Reporting Missingness	Medium	High	Inability to separate biogenic fuel CO□.	Documented that 81.4% (144/177) of EEA records lack excluding-biomass CO□; standardized on total direct CO_2 as verified regulatory proxy.
Data Leakage in Supervised Models	Critical	Low	Artificially inflated benchmark metrics.	Implemented 5-fold `GroupKFold` cross-validation grouped strictly by `facility_id`, completely isolating facilities between folds.

6.2 Technical Extension Roadmap: Multi-Sensor Operational Activity Proxying (VeriCBAM 2.0)

While the baseline proposal designs an initial Sentinel-5P tropospheric NO_2 operational presence check, extending the framework to multi-sensor remote sensing provides a compelling technical roadmap to independently reconstruct factory activity:

1. Thermal Infrared (TIR) Operating Timelines:

Industrial rotary cement kilns operate at internal temperatures exceeding 1,400°C, radiating external shell heat at 200°C–350°C, while primary steel tapholes pour molten iron at 1,500°C. Leveraging satellite thermal infrared sensors (such as Landsat-8/9 TIRS or high-resolution commercial thermal platforms) enables detecting continuous thermal anomalies, reconstructing an indisputable operational timeline of kiln firing versus maintenance shutdowns.

2. All-Weather Synthetic Aperture Radar (SAR):

Optical and thermal sensors remain constrained by cloud decks in temperate industrial zones. In contrast, Sentinel-1 C-band SAR penetrates persistent cloud cover, rain, and operates regardless of daylight. By tracking temporal changes in radar backscatter over heavy metal structures, scrap yards, and processing units, SAR establishes an all-weather activity index.

3. Computer Vision for Freight & Logistics Tracking:

Applying object detection to high-resolution optical imagery (Sentinel-2, PlanetScope) to monitor raw material stockpile depletion (limestone, coal, iron ore) and count bulk cargo barges and railcars at industrial terminals allows independent estimation of physical production volume (Q). This estimated volume can then be directly paired with stoichiometric intensity bounds.

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7. Deliverables & Resource Management

7.1 Core Software & Academic Deliverables

1. VeriCBAM Core Engine (`src/`): Clean, modular Python package featuring stoichiometric calculators (`cement_models.py`, `steel_models.py`), leave-one-year-out ratio extractors (`nox_baseline.py`), and Bayesian log-odds fusion (`evidence_fusion.py`).
2. Interactive 8-Tab Streamlit Decision Cockpit (`src/ui/app.py`):

- Tab 1: Executive Overview & Architecture: System KPIs, regulatory timeline, and 4-level evidence hierarchy.
- Tab 2: Facility Cohort & Geospatial Inspector: Interactive map highlighting selected plants with historical CO_2 timelines and legal threshold overlays.
- Tab 3: Stoichiometric Feasibility & Route Sandbox: Chemical mass-balance simulator testing calcination, iron reduction, and scrap dilution.
- Tab 4: Standalone Supplier Plausibility Pre-Check: Zero-history pre-screening module for importers and non-EU suppliers.
- Tab 5: Copernicus Sentinel-5P Satellite Inspector: Orbital NO_2 plume contrast and cloud-gating inspector.
- Tab 6: Bayesian Evidence Fusion & Audit Dossier: Waterfall log-odds chart, prior sensitivity curve, and 1-click Markdown audit dossier export.
- Tab 7: Empirical Benchmark Evaluation & Ablation: Live comparison of all 6 models across 325 graded scenarios with triage simulation.
- Tab 8: Capstone Proposal, References & Roadmap: Master academic documentation and direct EUR-Lex ELI links.

3. Automated Master Reproduction Pipeline (`scripts/reproduce_all.py`):

An end-to-end automated pipeline executing data ingestion, modeling, calibration, and benchmark evaluation stages with deterministic reproducibility.

4. Automated Verification Test Suite: Modular unit and integration test suite in `tests/` (`pytest`) validating software correctness, stoichiometry bounds, and boundary rules.
5. Open-Source GitHub Repository: Hosted at https://github.com/CaceresCristhian/VeriCBAM.

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8. Required Milestone Deliverables: LinkedIn Thought-Leadership Post

(Required under Milestone 2 — October 26, 2026)

Conceptual Project Announcement (Idea-Only, Zero Numerical Data)

A purely conceptual statement presenting the project initiative, the regulatory dilemma, and the evidence-fusion architecture without empirical numbers, statistics, or prematurely claimed results.

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With the European Union's Carbon Border Adjustment Mechanism (CBAM) entering definitive financial
enforcement, European importers of carbon-intensive commodities like steel and cement face legal
responsibility for the emissions embedded in their global supply chains.

The compliance challenge is immediate: verifying supplier carbon declarations requires physical on-site
audits, yet accredited verifiers are scarce worldwide and international site visits are logistically
constrained. Meanwhile, self-reported carbon spreadsheets cannot always be taken at face value.

For my Master’s Capstone in Data Science at the University of Europe for Applied Sciences (Potsdam),
supervised by Dr. Humera Noor, I am working on Project VeriCBAM: an explainable decision-support
framework designed to screen industrial emissions reports against independent evidence before physical
audits are scheduled.

Instead of treating emissions estimation as a black-box machine learning problem, the framework
approaches verification as an evidence-fusion challenge:
• Grounding declarations in first-principles chemical engineering (establishing mass-balance lower
bounds for iron reduction and limestone calcination).
• Cross-referencing facility reporting history and co-pollutant combustion profiles from independent
environmental registries.
• Leveraging spaceborne Earth observation as an external activity check to confirm whether industrial
sites were operational.
• Unifying these heterogeneous signals into a transparent, white-box decision model that helps compliance
officers prioritize where in-depth audits are truly needed.

The goal is not to replace accredited human verifiers, but to provide them with an objective,
physics-informed screening copilot that highlights physical inconsistencies and directs attention where
it matters most.

How is your organization approaching data integrity and verification in global carbon supply chains?

#DataScience #MachineLearning #PhysicsInformedAI #CBAM #IndustrialDecarbonization #SupplyChainCompliance
#ExplainableAI #CleanTech
```

9. Academic & Regulatory Reference Bibliography

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